

BPM 1

FTM: Lecture 39

Autonomic Integration ***Flipped Classroom***

Subramanya Upadhyaya, MD.

Professor

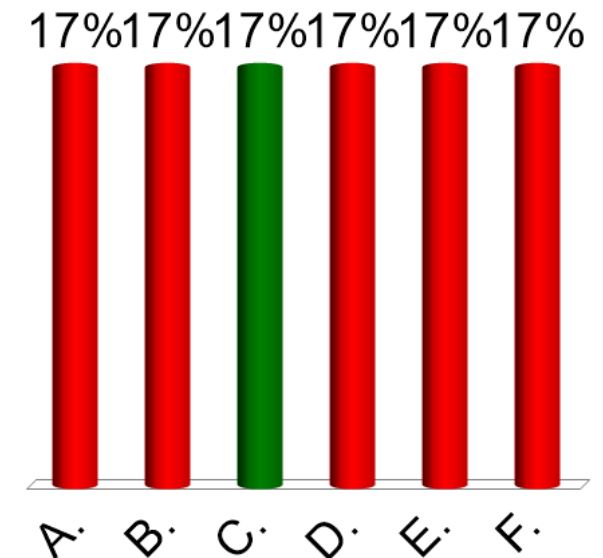
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A 54-year-old man comes to the physician because of a 6-week history of 2 episodes of chest pain. He had pain while climbing the stairs. The pain lasted for about 5 minutes and relieved by rest. He has a 6-year history of hypertension and is non-adherent with anti-hypertensive medications. His pulse is 88/min, blood pressure is 164/92 mm Hg, and respirations are 16/min. Physical examination shows loud second heart sound in aortic area. The physician prescribes him a medication that selectively blocks action of the sympathetic nerves to the heart. Which of the following autonomic receptors are most likely blocked by this drug?

- A. Alpha-1 receptors
- B. Alpha-2 receptors
- ✓ C. Beta-1 receptors
- D. Beta-2 receptors
- E. M1 muscarinic receptors
- F. M2 muscarinic receptors



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REQUIRED PRE-READING MATERIAL FOR FLIPPED CLASSROOM L39

- ❖ You are required to watch the video before attending L39 with the below link:

<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=c99731bf-c699-41e4-8264-b34e00d4756a>

- ❖ Read related material in the recommended Physiology textbook, chapter 6-Autonomic Nervous System, Under heading, “AUTONOMIC INTEGRATION”. Use link below to reach here:

<https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253086161&bookid=32>

LEARNING OBJECTIVES

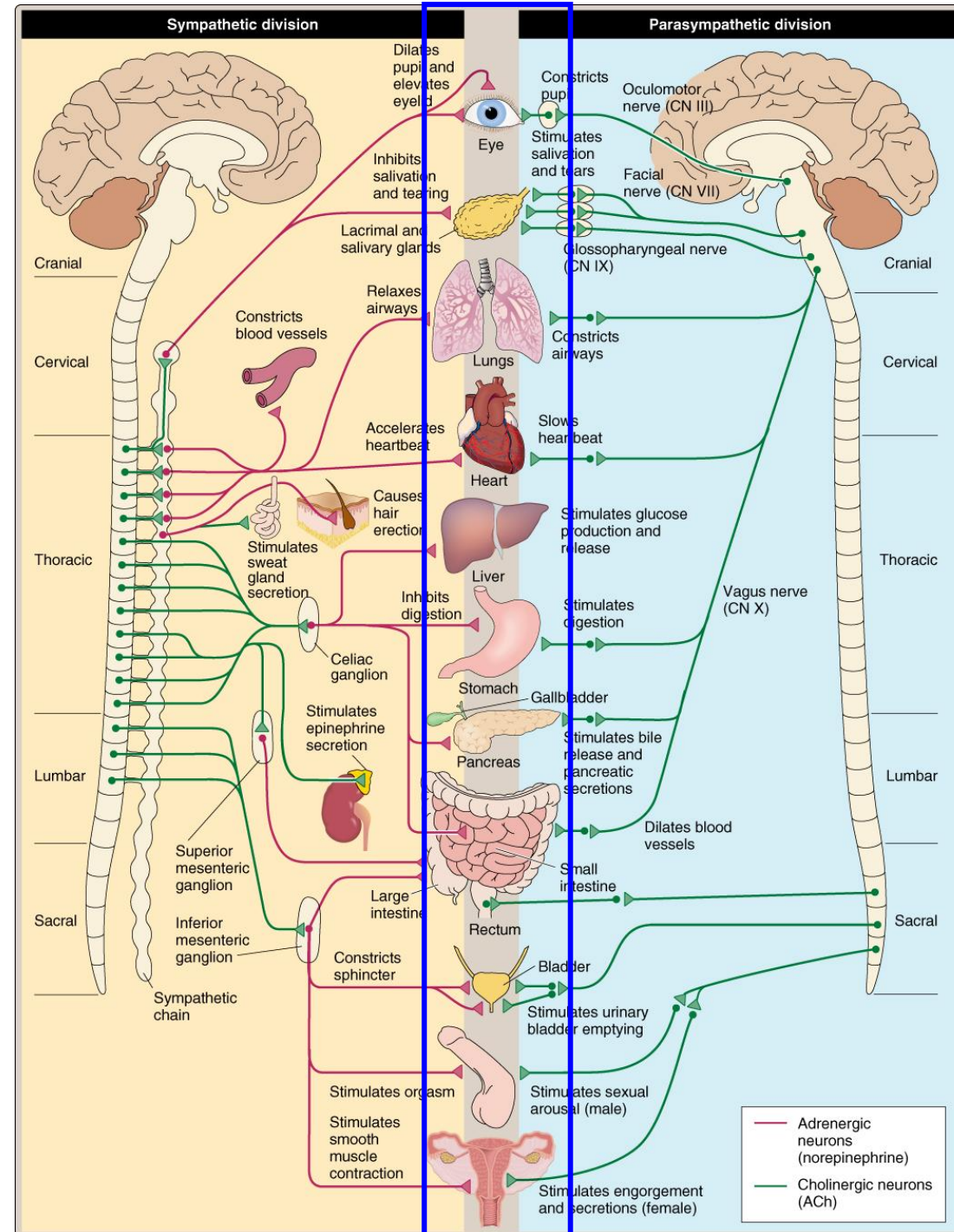
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0046** Explain the responses of end organs to stimulation of sympathetic and parasympathetic divisions of the autonomic nervous system.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0047** List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0048** List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

Lecture Outline

1. End organ responses to sympathetic and parasympathetic divisions
2. Autonomic functions of the eye
3. Baroreflex: components, functions, baroreflex dysfunction

Dual innervation and antagonistic effects of ANS

There is an
equilibrium between
sympathetic and
parasympathetic
effects



List target organs that
has **only sympathetic**
innervation:

Modified from Bear, M.F.,
Connors, B.W., and Paradiso,
M.A. Neuroscience—Exploring
the Brain. Third Edition.
Lippincott Williams & Wilkins,
2007

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Skin	Sweat glands	None	Secretion (cholinergic muscarinic receptors)
	Arrector pili muscles	None	Contraction α_1
Blood vessels	Skin (general)	None	Constriction α_1
	Skeletal muscle	None	Dilation β_2
	Viscera	None	Constriction α_1
Heart	SA Node	Decrease heart rate M2	Increase heart rate β_1
	AV Node	Decrease conduction velocity M2	Increase conduction velocity β_1
	Atria	Decrease contractility M2	Increase contractility β_1
	Ventricle	None	Increase contractility β_1

End organ responses to sympathetic and parasympathetic divisions

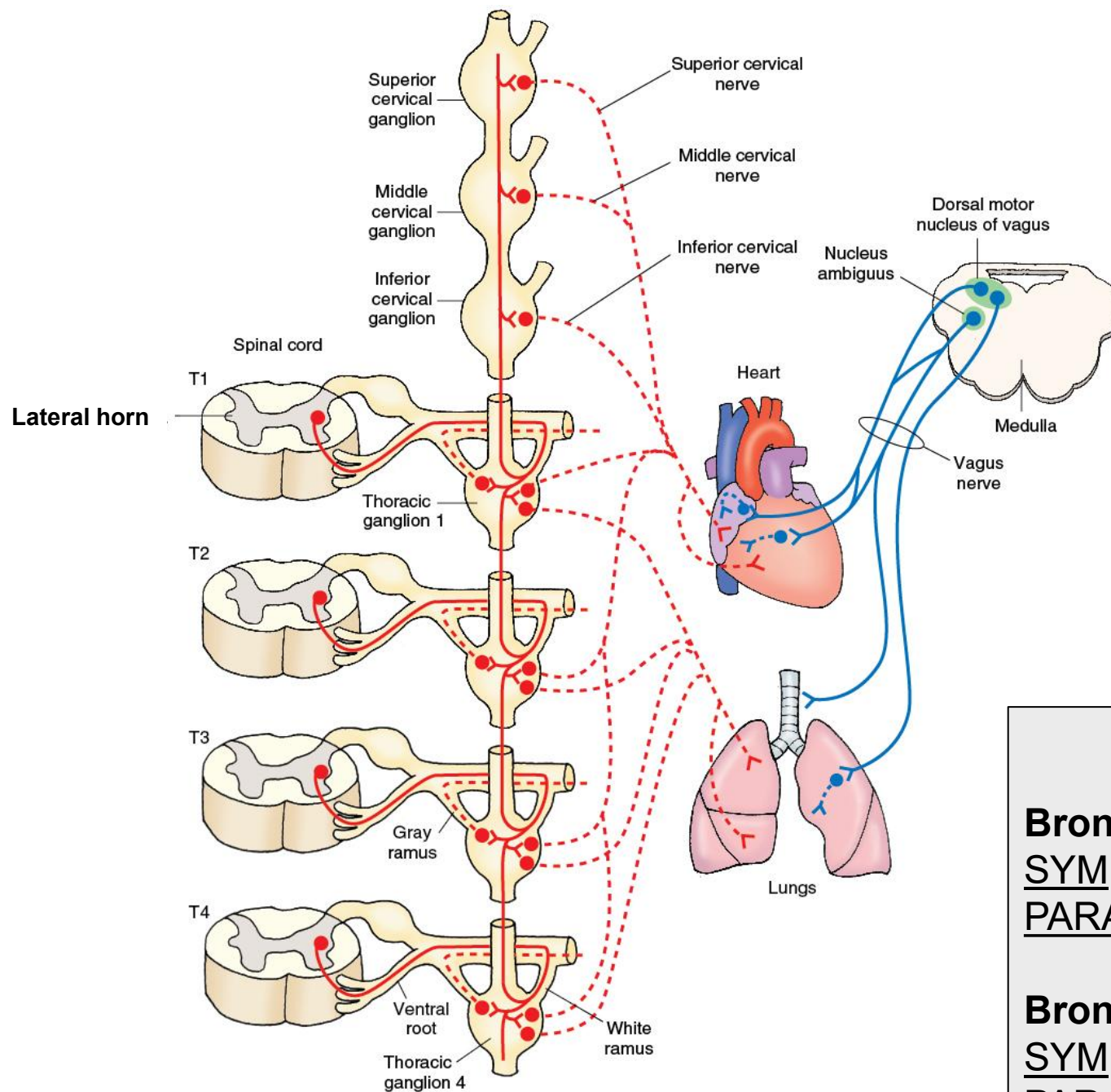
Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Lungs	Bronchioles	Constriction	Dilation β_2
	Glands	Secretion	Decrease
Gastrointestinal tract	Wall muscles	Contraction M3	Relaxation
	Sphincters	Relaxation	Contraction
	Glands	Stimulation M3	Inhibition
Metabolic functions	Liver	None	Glycogen breakdown and new glucose production
	Pancreas (insulin)	Increased secretion	Decreased secretion α_2
	Adipose Cells	None	Lipolysis β_3
	Kidney	None	Renin release β_1
	Adrenal medulla	None	Secretion of catecholamines (cholinergic nicotinic)

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Urinary system	Detrusor	Contraction M3	Relaxation
	Sphincter	Relaxation	Contraction $\alpha 1$
Reproductive system	Uterus	Variable	Relaxation β_2
	Genitalia	Erection (ACh & NO) M3	Ejaculation
Eye	Circular muscle	Contraction M3	<i>None</i>
	Ciliary muscle	Contraction M3	<i>None</i>
	Radial muscle	None	<i>Contraction $\alpha 1$</i>
	Ciliary epithelium	None	<i>Secretion of aqueous humor $\beta_{1,2}$</i>
	Eyelid muscle (superior tarsal muscle)	None	<i>Elevates upper eyelid</i>

Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

HEART & LUNGS



HEART

Sinoatrial node: Heart Rate

SYM: increases β_1

PARASYM: decreases M_2

Atrioventricular node:

Conduction velocity

SYM: increases β_1

PARASYM: decreases M_2

Ventricular muscle

SYM: increases contractility β_1

LUNGS

Bronchial smooth muscle

SYM: bronchodilation β_2

PARASYM: bronchoconstriction M_3

Bronchial glands

SYM: decrease secretion β_2

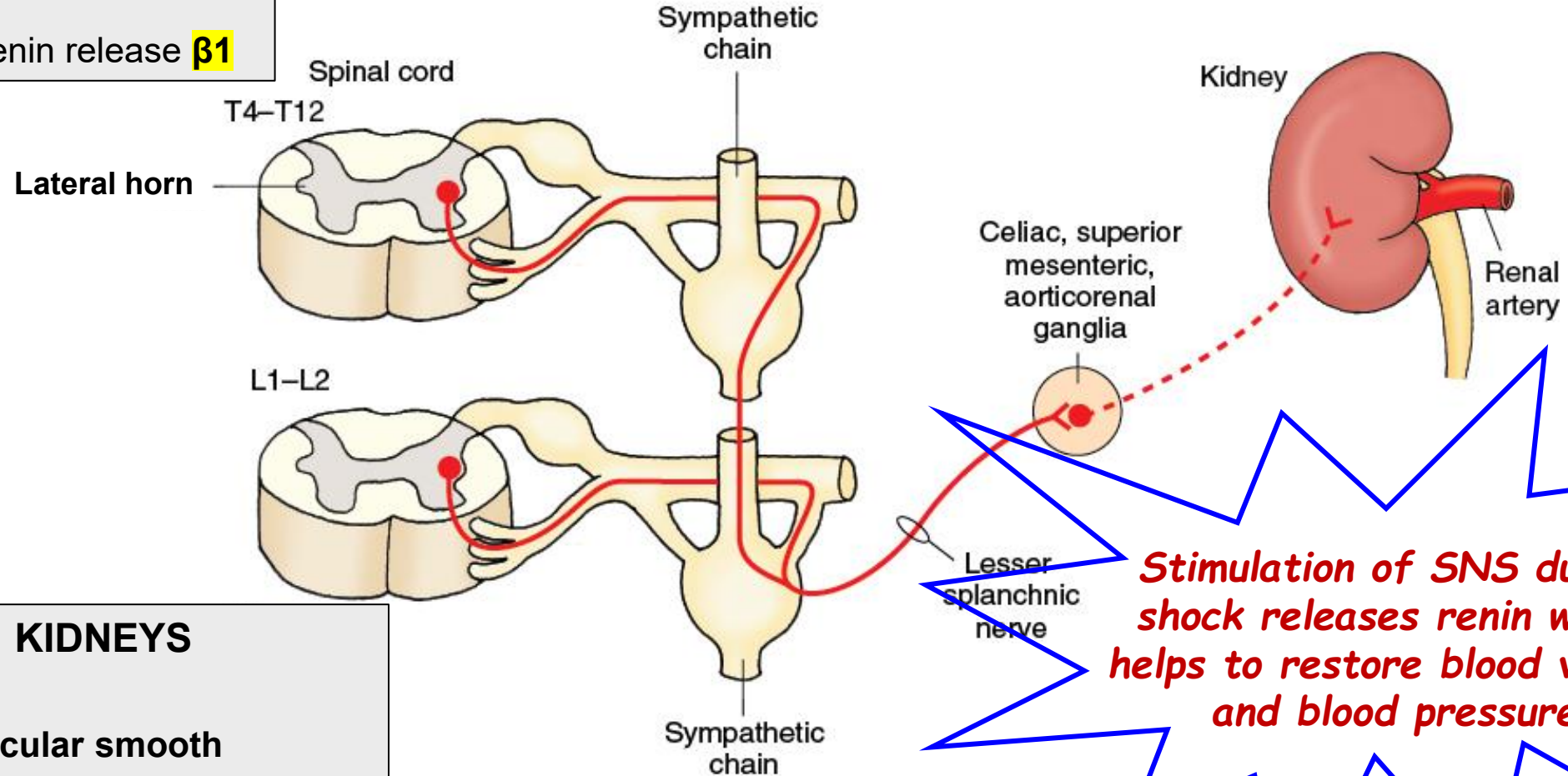
PARASYM: increase secretion M_3

KIDNEYS

KIDNEYS

Juxtaglomerular cells:

SYM: ↑ Renin release **β1**



KIDNEYS

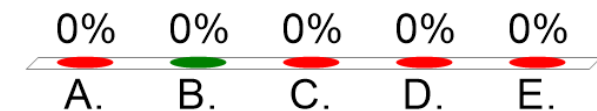
Renal vascular smooth muscles:

SYM: Relax smooth muscles **D1**

CLICKER INTERACTIVE QUESTIONS NEXT!

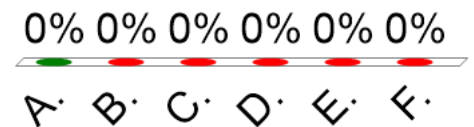
A 62-year-old woman comes to the physician because of a 2-week history of palpitations and shortness of breath. She has a 15-year history of type 2 diabetes mellitus. Her vital signs are within normal limits. Physical examination shows irregularly irregular pulse. An ECG shows absent P waves and irregular R-R intervals. She is diagnosed with atrial fibrillation. She is prescribed a drug that inhibits adrenergic receptors in the heart. Which of the following cellular alterations in patient's heart is most likely after taking this drug?

- A. Increase in adenylyl cyclase activity
- ✓ B. Decrease in adenylyl cyclase activity
- C. Increase in phospholipase C activity
- D. Decrease in IP₃, DAG, and Ca⁺⁺
- E. Increase in nitric oxide

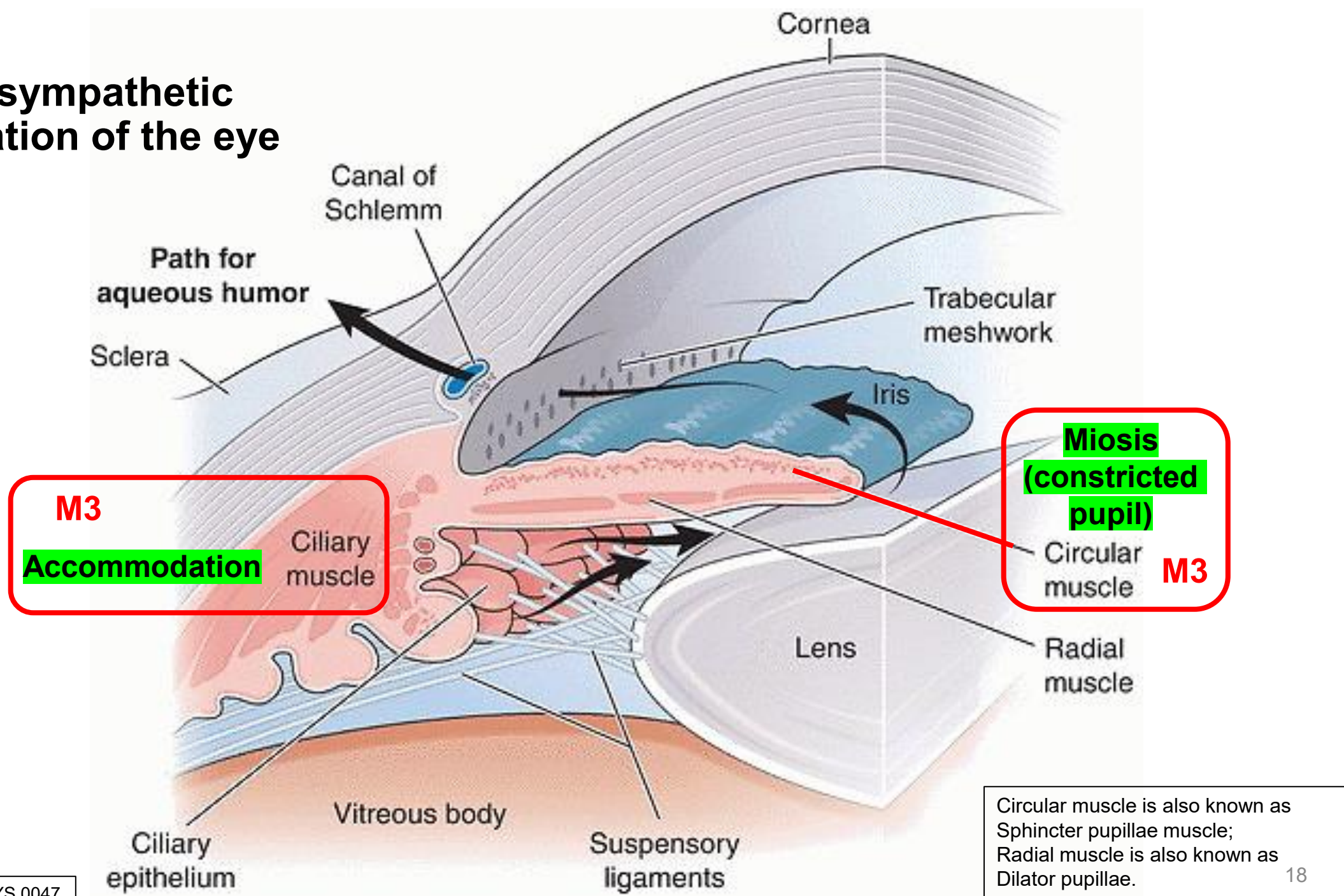


A 56-year-old man is brought to the emergency department because of a 15-minute history of severe headache and altered mental state. His wife says he was working in the yard when the event occurred. He has a 10-year history of hypertension and is not adherent to anti-hypertensive medications recently. His blood pressure is 190/102 mm Hg. Neurologic examination shows neck stiffness, altered consciousness, and dilated left pupil which is not reacting to light. He is diagnosed with brain hemorrhage and compression of a cranial nerve that carries parasympathetic nerves to the eyes. Which of the following nerves is most likely compressed in this patient?

- ✓ A. III cranial nerve
- B. VII cranial nerve
- C. Cervical sympathetic nerves
- D. IX cranial nerve
- E. X cranial nerve
- F. Thoracic sympathetic nerves



Parasympathetic Innervation of the eye



A 45-year-old man comes to the physician because of a 6-month history of blurred vision. He says he cannot read small prints in the newspaper. His vital signs are within normal limits. Physical examination shows difficulty to read letters in the near-vision card. He is diagnosed with presbyopia (inability to focus nearby objects) because of weakness of the ciliary muscles. Parasympathetic division of the autonomic nervous system regulates the contraction of these muscles. Which of the following types of autonomic receptors are most likely involved in this process?

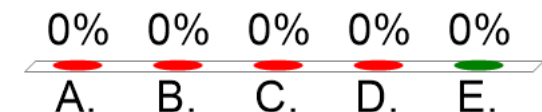
A. Alpha-1

B. Beta-1

C. M1

D. M2

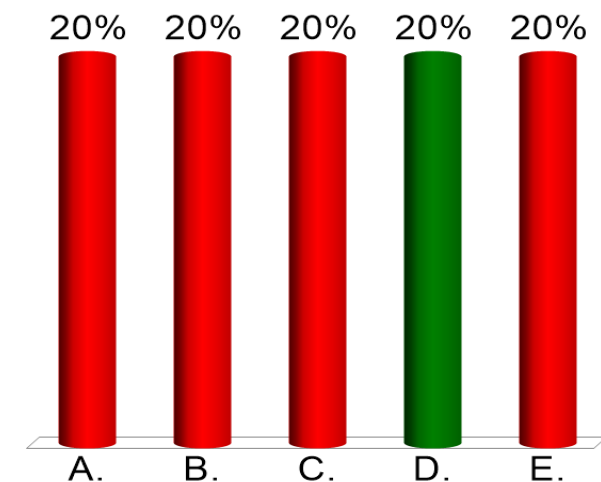
✓ E. M3



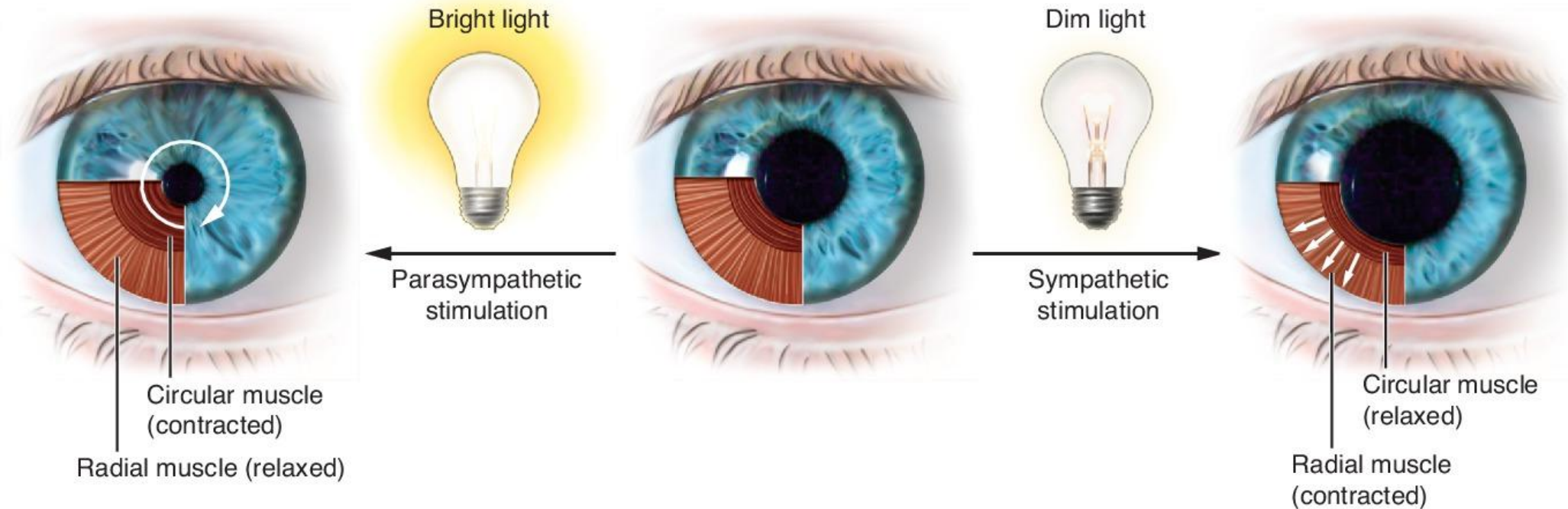
A 73-year-old man comes to the physician for a follow-up examination. He has a 15-year history of type 2 diabetes mellitus. His vital signs are within normal limits. Physician performs a pupillary light reflex in a dim-light room. When his left eye is illuminated, both pupils constrict. Which of the following second messengers is most likely associated with this physiologic eye response in the patient?

- A. cAMP
- B. Protein kinase A
- C. cGMP
- ✓ D. IP3
- E. Tyrosine kinase

https://th.bing.com/th/id/R.348bc7490fdb5d89eefab0acffc92c9?rik=4DpEbGR8oNnwrA&riu=http%3a%2f%2fi.makeagif.com%2fmedia%2f2-04-2016%2fO_sRdF.gif&ehk=jECOeNSCexKfQ%2bFILi4VvjVHz%2fV7Nv4c5fi3FRDxhIA%3d&risl=&pid=ImgRaw&r=0



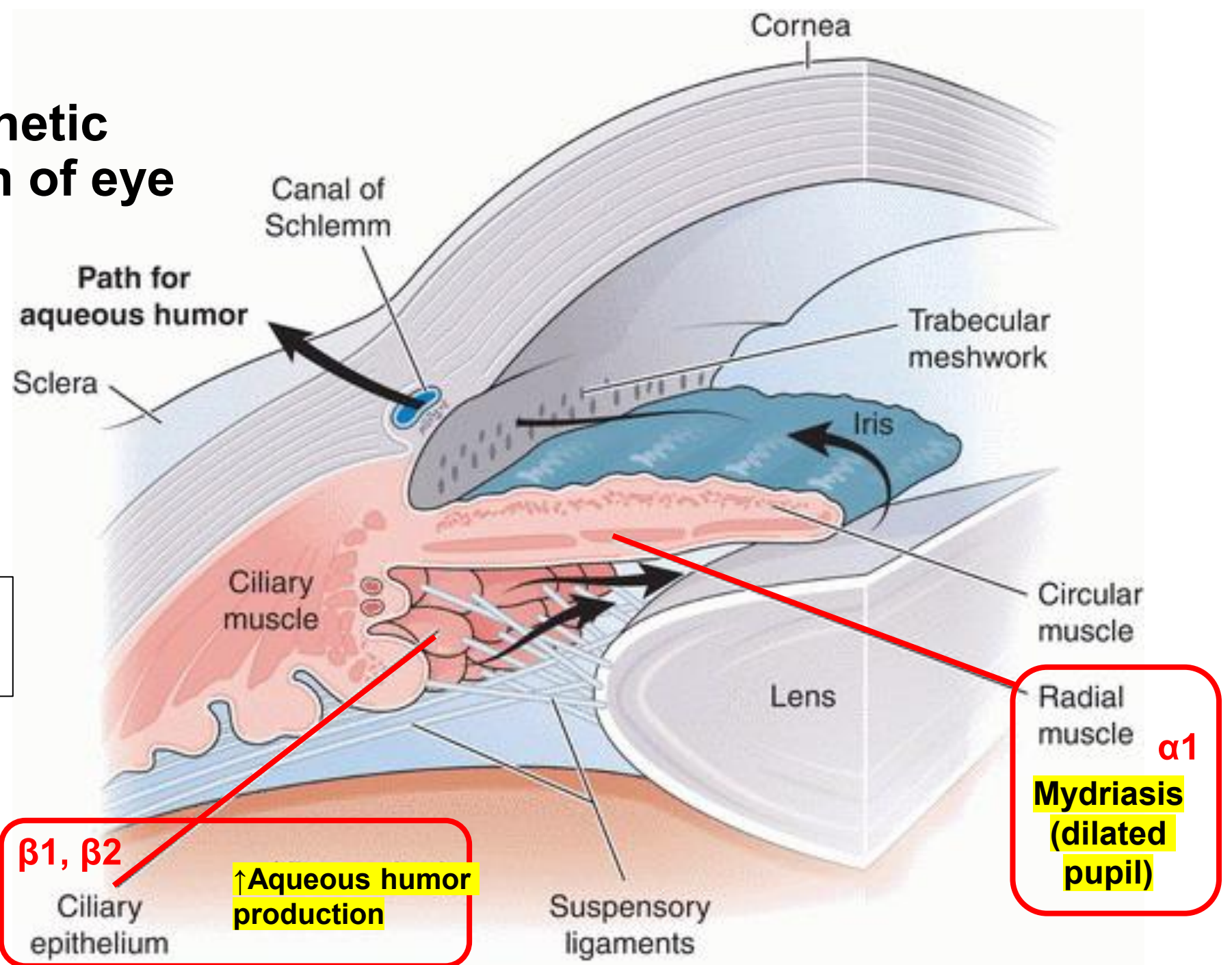
Autonomic regulation of pupil light adaptation



Pupil size adapts to changing light conditions

Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

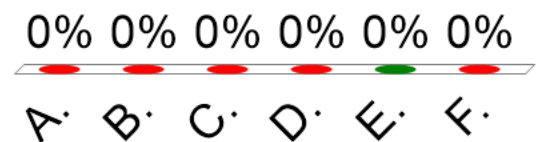
Sympathetic innervation of eye



Circular muscle is also known as Sphincter pupillae muscle; Radial muscle is also known as Dilator pupillae.

A 76-year-old woman comes to the physician because of a 1-day history of pain and blurry vision in her right eye. She has a 10-year history of hypertension. Her pulse is 88/min and blood pressure is 138/90 mm Hg. Physical examination of the eye shows a reddish and inflamed sclera in the right eye. Tonometry shows elevated intraocular pressure. She is diagnosed with glaucoma and prescribed a beta-blocker containing eyedrop to reduce the rate of formation of the aqueous humor. Function of which of the following structures in the eye is most likely inhibited by this medication?

- A. Dilator pupillae muscle (radial muscle)
- B. Sphincter papillae muscle (circular muscle)
- C. Ciliary muscle
- D. Canal of Schlemm
- ✓ E. Ciliary epithelium
- F. Suspensory ligaments

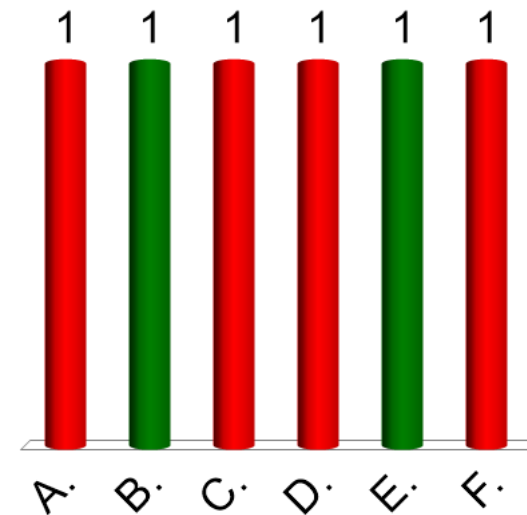


A 55-year-old man comes to the physician because of a 4-week history of dry cough and shortness of breath. He has smoked 2 packs of cigarette daily for 26 years. Physical examination shows miosis and drooping (ptosis) of the right eye. The right forehead is dry. An MRI of the chest shows a mass in the right upper lobe (see image). He is diagnosed with a lung tumor that compresses and damages the right cervical sympathetic ganglia. Defective functioning of which of the following structures in the right eye are most likely in this patient? (choose **more than one answer**)



MRI chest

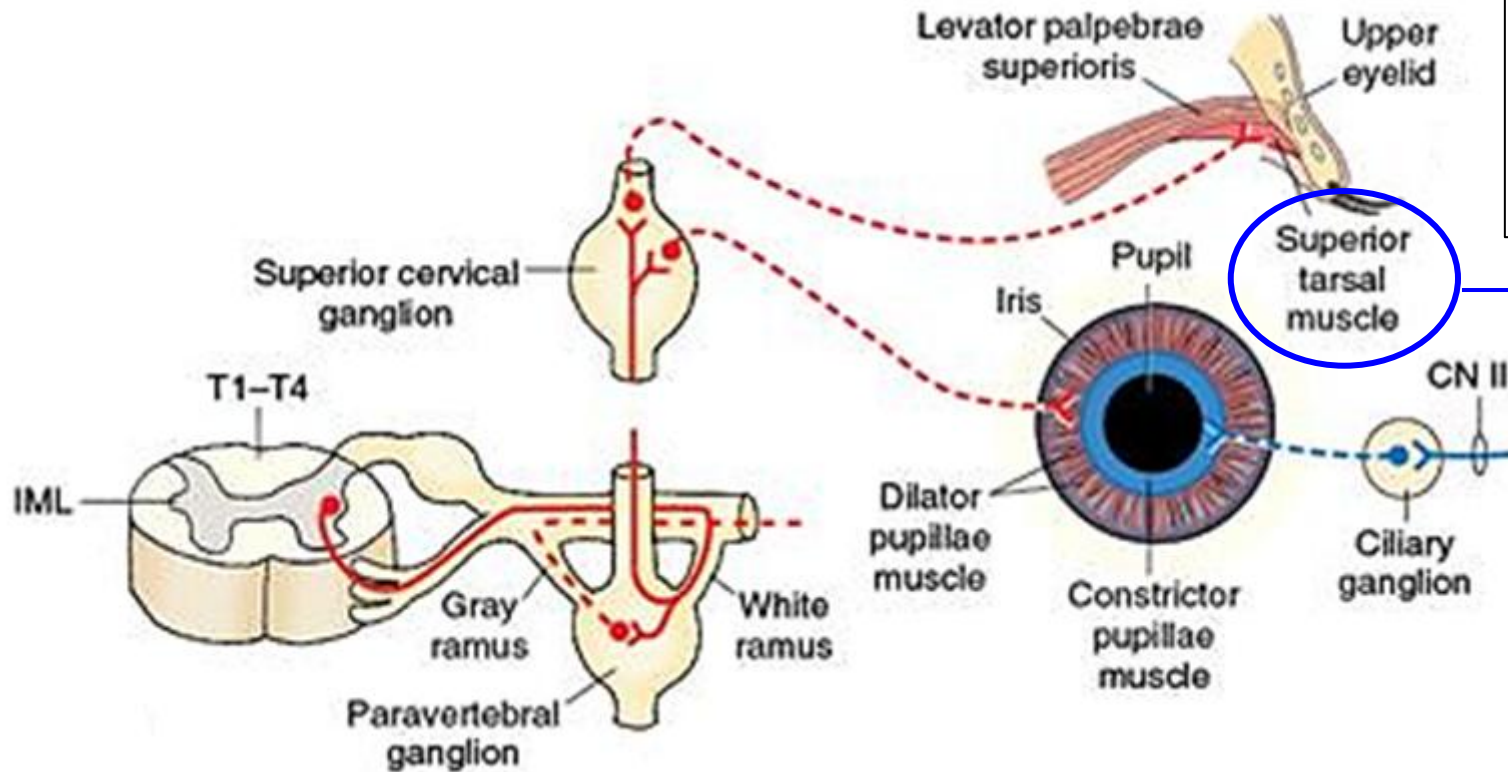
- A. Ciliary muscle
- ✓ B. Superior tarsal muscle
- C. Levator palpebrae muscle
- D. Sphincter pupillae muscle
- ✓ E. Dilator pupillae muscle
- F. Orbicularis oculi muscle



Horner syndrome



Autonomic innervation of the face and eye: Supplied by the superior cervical sympathetic nerves



Eyelid elevation is by contraction of 2 muscles:

- Levator palpebrae superioris
- Superior tarsal muscle**-innervated by cervical sympathetic nerves



Patient with hyperthyroidism

Staring look in hyperthyroid patient is partly because of contraction of this muscle!

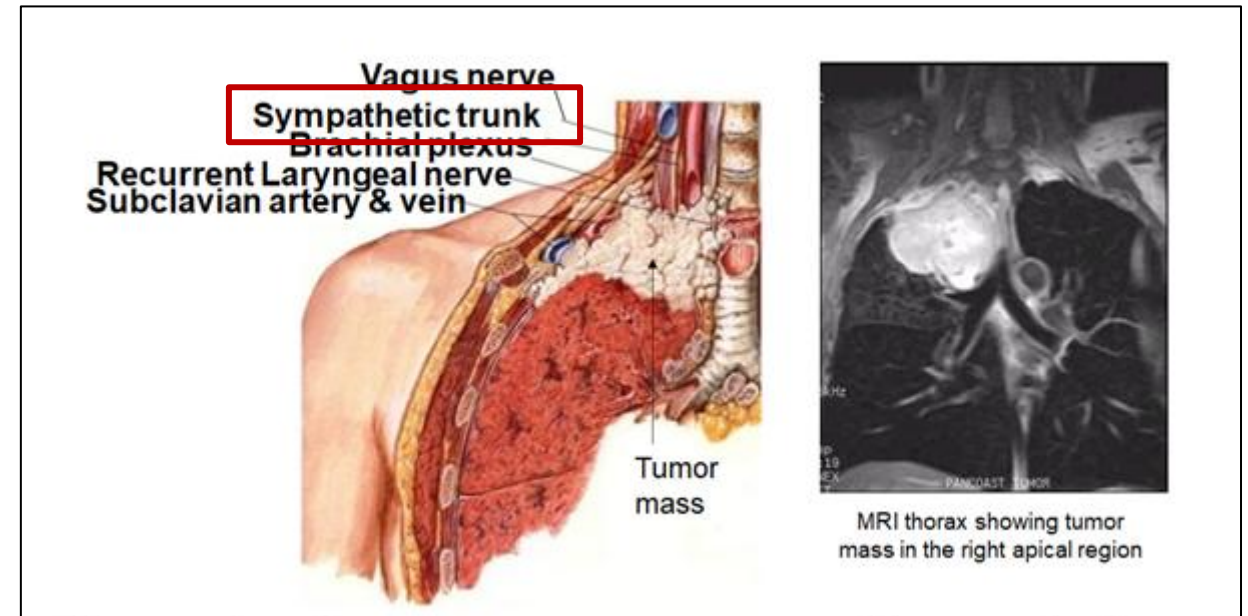
Horner syndrome: Sympathetic denervation of the face because of a tumor in the apical region of the lung

A tumor in the apical region causes damage to the **cervical** sympathetic ganglia

Symptoms:

On the side of the face that is supplied by the damaged nerve

- Ptosis (drooping) of the upper eyelid
- Anhidrosis (absence of sweating) in the face
- Miosis (pupil constriction)



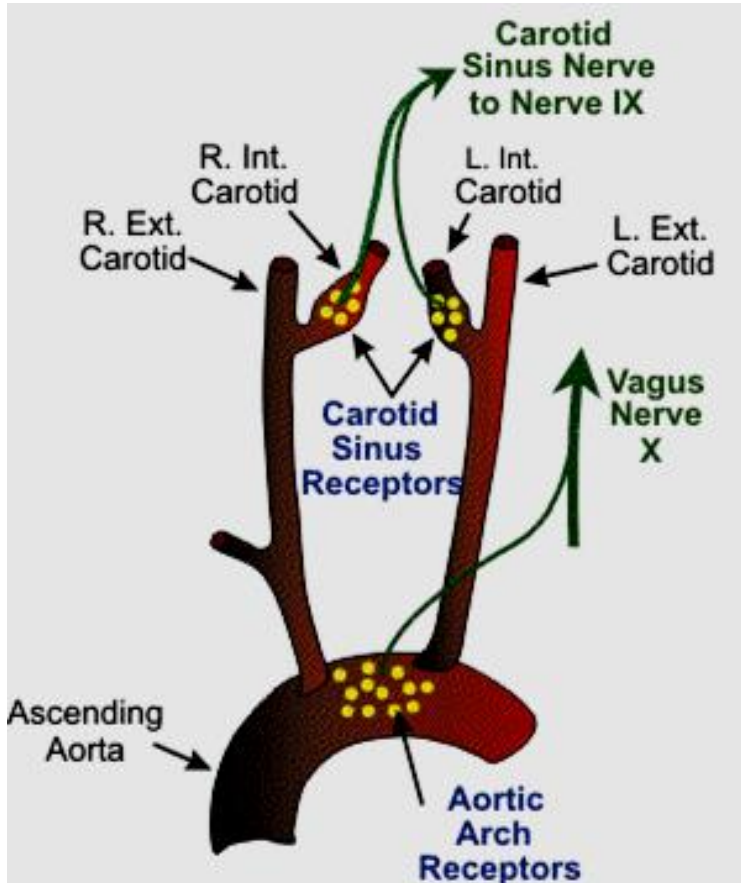
Affected side of face . Normal side of face

A 72-year-old man comes to the physician because of a 6-month history of lightheadedness and visual blurring while standing off the bed. He has a 10-year history of Parkinson disease and is taking medications. His blood pressure in supine is 132/84 mm Hg and in standing is 110/70 mm Hg. He is suspected to have a dysfunction of the baroreflex. Which of the following structures most likely senses change in blood pressure while standing off the bed in this patient?

- A. Vagus nerve to the heart 0
- ✓ B. Baroreceptors in major arteries 0
- C. Sympathetic nerve to the heart 0
- D. Mechanoreceptors in the skin 0
- E. Sensory receptors in weight-bearing joints 0



Arterial Baroreceptors initiate Baroreflex to regulate BP



Increase in Blood Pressure (**stimulus**)



Stretching of baroreceptors (**Sensors**)



Increased action potentials in the afferent nerves connected to the baroreceptors (**Afferent n**)

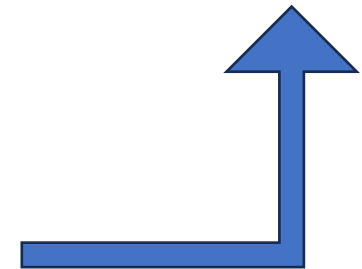


Cardiovascular regulatory centers in brain stem (**Center**)

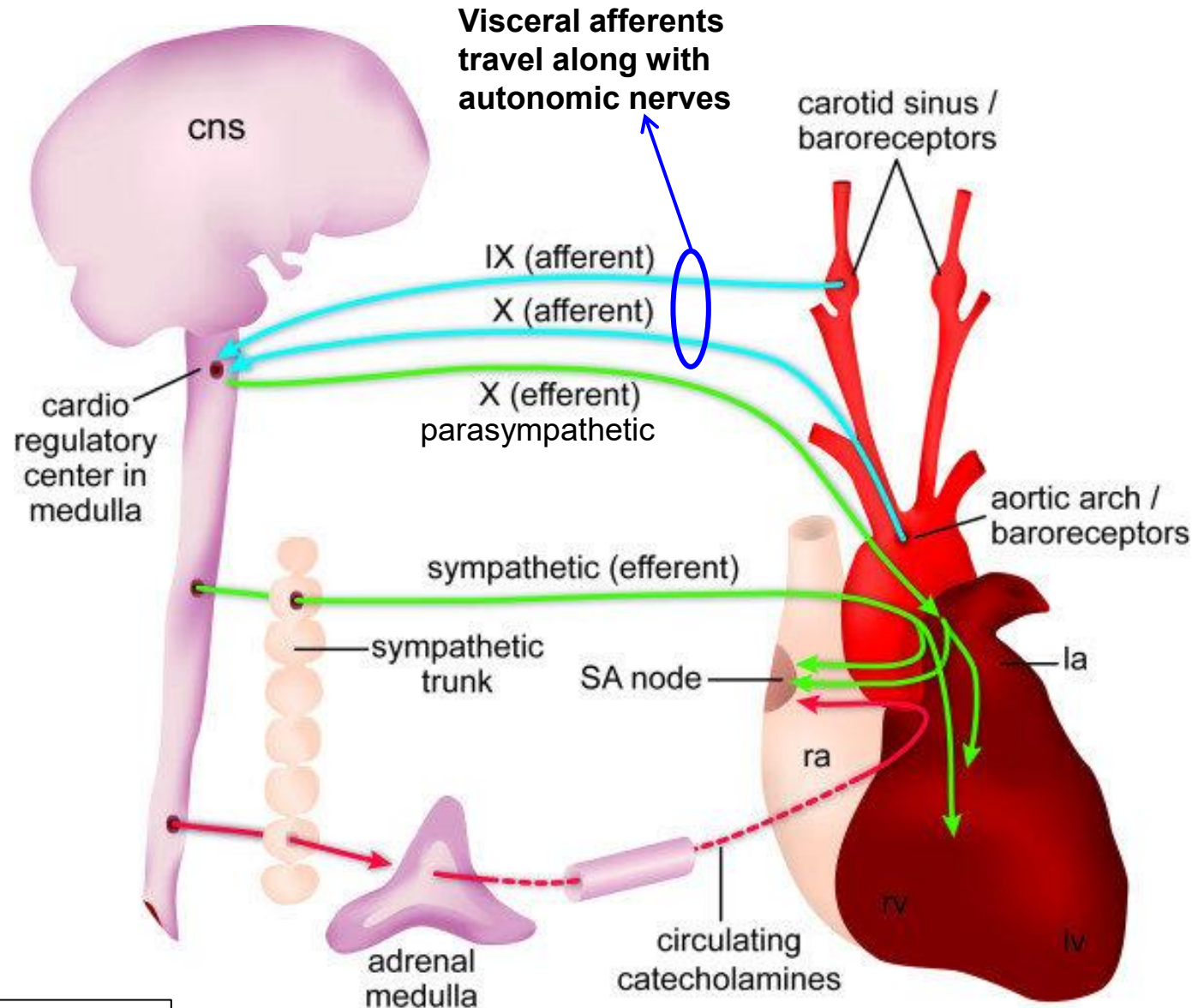


Influence on Autonomic NS (**Efferent n**)

BP back to normal (**Effect**)



Arterial Baroreceptors regulate blood pressure (BP)

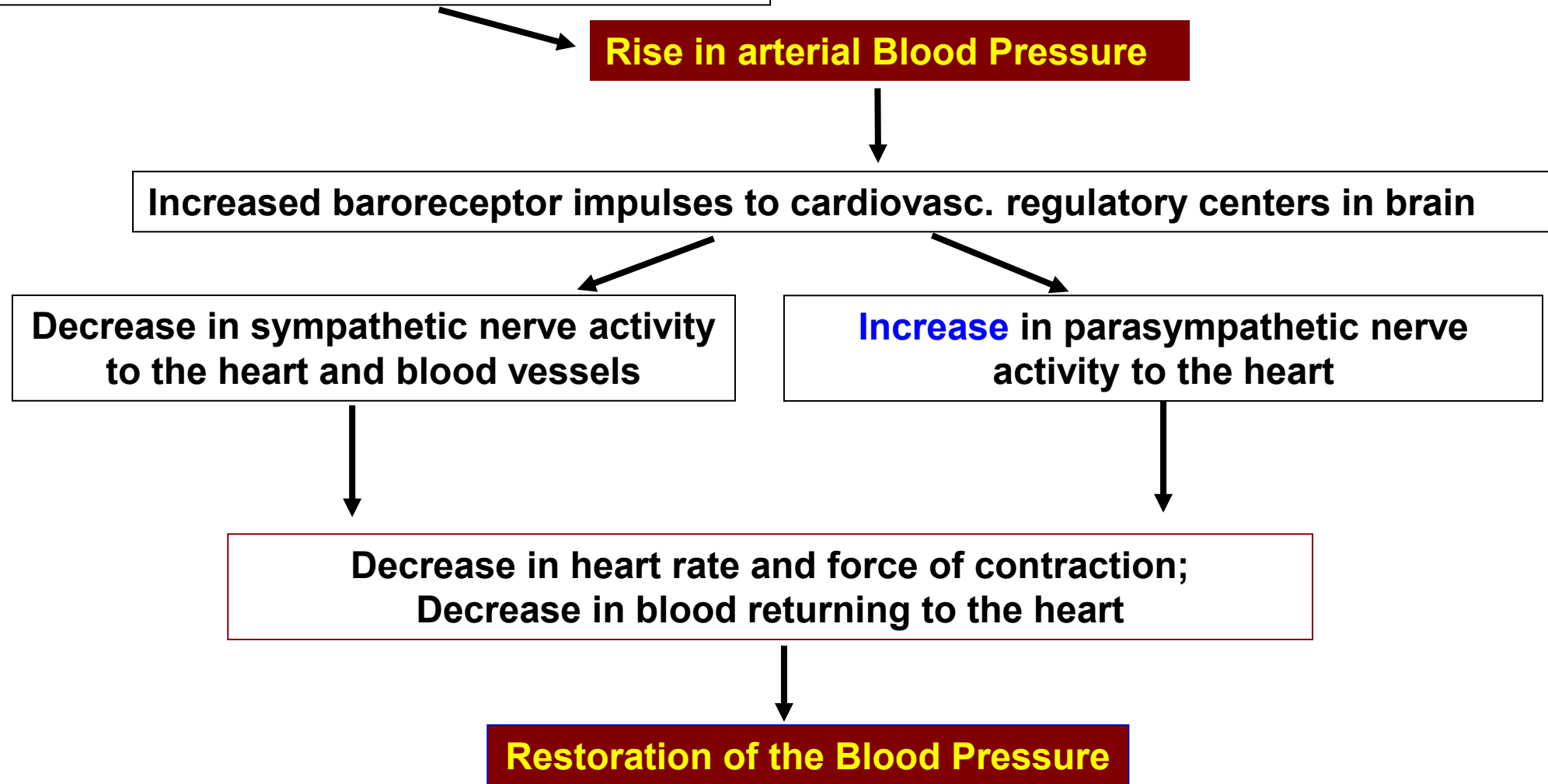


Baroreceptors
are sensors
located in carotid
sinus and aortic
arch

Blue: Afferent
nerves;
Green: Efferent
nerves;
Red: Efferent nerve
supplying adrenal
medulla

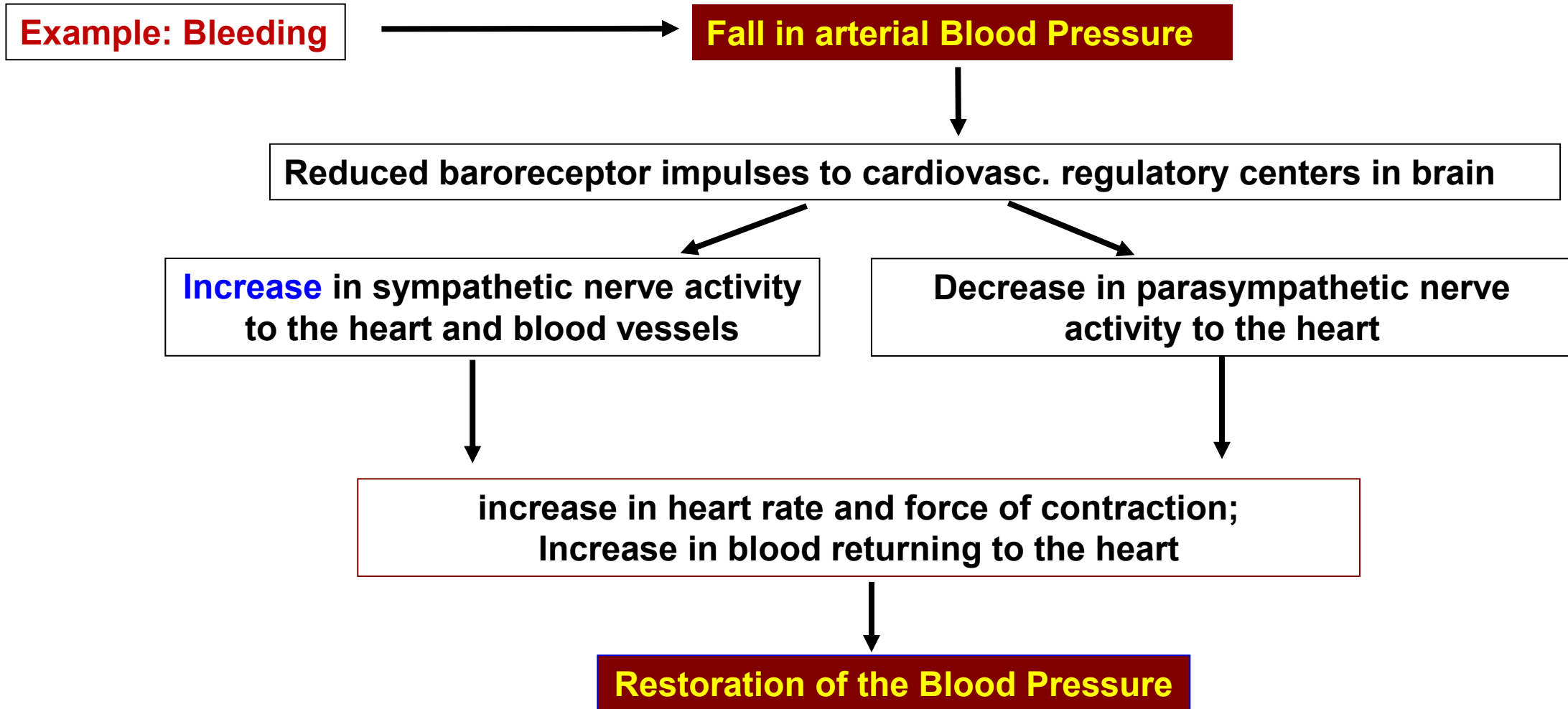
Baroreflex response to a sudden rise in BP

Example: Change posture from standing to supine



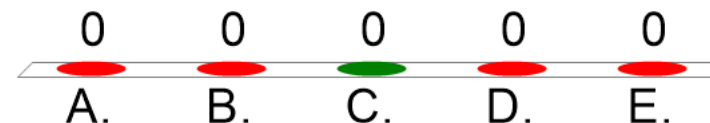
Task: Draw a similar flow chart to show baroreflex response to a **sudden decrease** in the Blood Pressure

Baroreflex response to a sudden lowering BP (Bleeding)



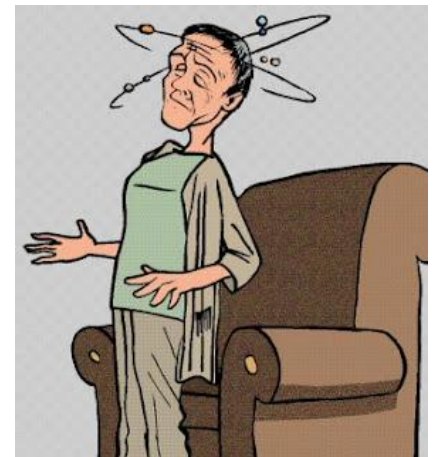
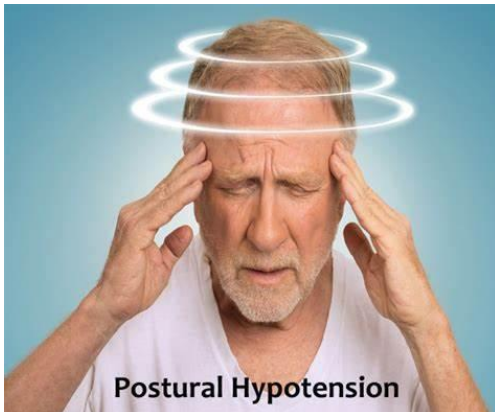
A 26-year-old man is brought to the emergency department 20- minute after he was involved in a motor vehicle collision. He was bleeding and has multiple fractures of his limbs. His pulse is 120/min, blood pressure is 90/62 mmHg. Physical examination shows altered consciousness, pallor, and profuse sweating. Sympathetic nerve activity is increased in his entire body. His skin is cold to touch because of vasoconstriction of the skin blood vessels. Which of the following autonomic receptors is most likely active in this patient's skin blood vessels?

- A. M1
- B. M3
- ✓ C. Alpha-1
- D. Alpha-2
- E. Beta-1



Why baroreflex is important in day-to-day life?

- ❖ Patient with dysfunction of the baroreflex is unable to maintain normal blood pressure under different circumstances.
- ❖ This leads to a low blood pressure while changing body posture from supine to erect and is called **orthostatic hypotension** (postural hypotension)
- ❖ Patient complaints:
 - Lightheadedness
 - Blurry vision
 - Vertigo/dizziness
 - Transient loss of consciousness (syncope)



Any Questions ?

How to reach me: Either walk into my office or email.

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Professor,

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Sciences, Lower Charter Hall,
School of Medicine, St. George's University.**

Email: supadhy@sgu.edu

Other Recommended readings

SIEGEL A, SAPRU HN (2019). Essential neuroscience. Philadelphia, Wolters Kluwer Health/Lippincott Williams & Wilkin

<https://meded.lwwhealthlibrary.com/content.aspx?bookId=2482§ionId=196989456&resultClick=1>

Moore KL, Dalley AF II; Agur AMR. Clinically Oriented Anatomy, 8e
<https://meded.lwwhealthlibrary.com/book.aspx?bookid=2212>

Le, Tao and Bhushan, Vikas. First Aid for the USMLE Step 1 2025, Thirty Fifth edition. New York: McGraw-Hill Education, 2025.